

NEUR 540 • GRADUATE SEMINAR • FALL 2026

# Computational Neuroscience: from spikes to circuits

A twelve-week graduate course modelling how neurons compute, individually and together.



# What this first session covers.

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- 01 What the course is about
  - 02 The twelve-week arc
  - 03 What you will be able to do by the end
  - 04 Logistics, grading, and what to do this week
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# This course builds a working brain model, one layer at a time.

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We start from a single neuron's electrical behaviour and end with networks that learn. Every week adds a layer: from the membrane equation, to spiking models, to populations, to plasticity. By December you will have coded each one yourself.



# Who this course is for, and what it assumes.

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This is a graduate seminar for students with calculus, basic linear algebra, and some Python. We assume no prior neuroscience — the biology is taught alongside the math. If you can integrate a differential equation and write a for-loop, you are ready.

- Prerequisites: calculus, linear algebra, intro Python.
- No prior neuroscience required.



TWELVE WEEKS • THREE MODULES

# How the course unfolds across the term.

| MODULE      | WEEKS 1–4            | WEEKS 5–8            | WEEKS 9–12            |
|-------------|----------------------|----------------------|-----------------------|
|             | SINGLE NEURON        | POPULATIONS          | LEARNING              |
| Theory      | ✓ Membrane & cable   | – Network dynamics   | ○ Synaptic plasticity |
| Coding labs | ✓ Integrate-and-fire | – Recurrent networks | ○ Learning rules      |
| Readings    | ✓ Hodgkin–Huxley     | – Attractor models   | ○ Predictive coding   |

✓ Shipped      – In flight      ○ Planned



# By the end of the course, you will be able to:

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01

## **Model a single neuron**

Derive and simulate the membrane equation and the integrate-and-fire model from first principles.

02

## **Build a network**

Implement recurrent populations and explain how circuit dynamics give rise to memory and decisions.

03

## **Implement learning**

Code biologically grounded plasticity rules and show how a network learns from experience.

04

## **Read the literature critically**

Evaluate a computational neuroscience paper's model, assumptions, and claims.



# How you are assessed: code more than exams.

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Grades come mostly from the weekly coding labs, where you build the models we derive in lecture. A midterm tests the theory, and a final project lets you model a system of your own choosing. There is no final exam.

- Weekly labs: 50%. Midterm: 20%. Final project: 30%.
- No final exam — the project is the capstone.



# The final project: model one phenomenon end to end.

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In the last month you will pick a neural phenomenon — a visual illusion, a decision, a memory effect — and build a computational model that reproduces it. You will present it in week twelve. Past students have modelled birdsong learning, place cells, and binocular rivalry.

- Pick a phenomenon by week eight; build it by week twelve.
- A short talk plus your code is the deliverable.



# Course policies, briefly.

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Labs are due Fridays at 5pm; two no-questions late days are yours to spend across the term. Collaboration on labs is encouraged, but the code and the write-up must be your own. All materials are open and posted to the course site.

- Two flexible late days, no explanation needed.
- Discuss freely; submit your own work.



# What to do before next week.

- ☒ Enroll and join the course forum.
- ☐ Install the Python environment from the setup guide.
- ☐ Read Chapter 1 — the membrane equation.
- ☐ Run the warm-up notebook and post one question.
- ☐ Skim the syllabus and flag any scheduling conflicts.



# Office hours and how to get unstuck.

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I hold office hours Tuesdays and Thursdays, and the TA runs a lab clinic before each deadline. The fastest help is the forum — post your error and someone, often a classmate, will answer within hours. Do not suffer alone with a bug.

- Office hours Tue/Thu; TA clinic before each lab is due.
- Post errors to the forum first — it is the fastest channel.



INSTRUCTOR • PROF. HANA YOSHIDA • COURSE SITE AT [NEUR540.EXAMPLE](#)

By December, you will have built  
a brain that learns — line by line.